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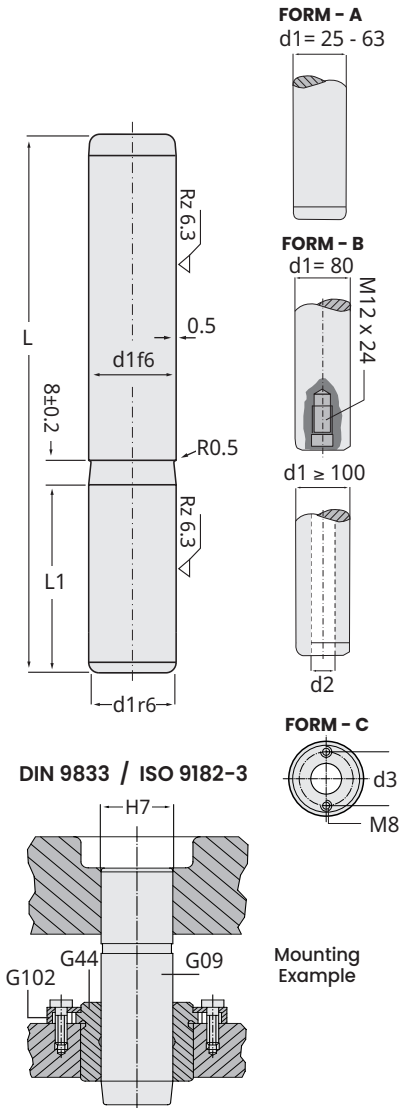
18

Guiding Components



Code: **G09**

Guide Pillar for Large Dies
DIN 9833



DIN 9833 / ISO 9182-3

Mounting
Example

Heavy Duty Type for Dies

Practises: Ground surface is holeless up to $\varnothing d = 80$ and $\varnothing 80 = M12 \times 24$ mm has one lifting hole. From $\varnothing d100$, it has hollow ($d2$) and have 2 pieces $M8 \times 24$ mm lifting holes.
Note: Hole tolerance should be H7.
It is recommended to use guide pillar only with self-lubricating guide bushes.
Compatible Bushes:
Self-lubricating bronze guide bushes.

d1	L	d2 Hole	d3	L1	Form
25	125	-	-	40	A
	140				
	160				
	180				
	200				
	224				
32	140	-	-	45	A
	160				
	180				
	200				
	224				
	250				
40	140	-	-	56	A
	160				
	180				
	200				
	224				
	250				
50	160	-	-	70	A
	180				
	200				
	224				
	250				
	280				
63	180	-	-	80	A
	200				
	224				
	250				
	280				
	315				

d1	L	d2 Hole	d3	L1	Form
80	200	-	-	100	B
	224				
	250				
	280				
	315				
	355				
100	400	50	72	125	C
	450				
	500				
	560				
	315				
	355				
125	400	65	90	140	C
	450				
	500				
	560				
	315				
	355				
160	400	95	132	180	C
	450				
	500				
	560				

Operating Components:
Self-lubricating guide bushes
are recommended.



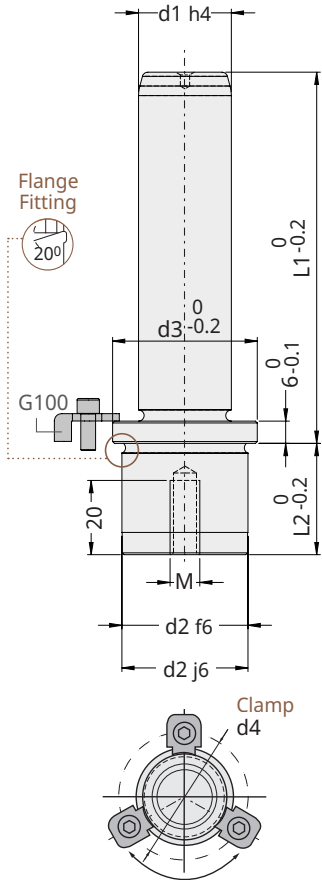
Order: **G09**. d1 x L



Material: 1.1213 (CF53)
Hardness: 58 - 62 HRC

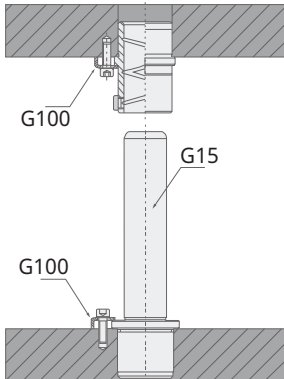


Guide Pillar, Thickbacked



Mounting Example

External diameter of bush and rear positioning diameter of the G15 guide pillar are same.



d1	L1	d2	d3	L2	d4 Clamp	M
20	100	28	34	23	46.2	M8
	125					
	140					
	160					
	180					
	200					
25	100	34	39	30	51.4	M8
	125					
	140					
	160					
	180					
	200					
30	220	39	44	37	56.6	M8
	240					
	260					
	280					
	300					
	320					
	340					
	360					

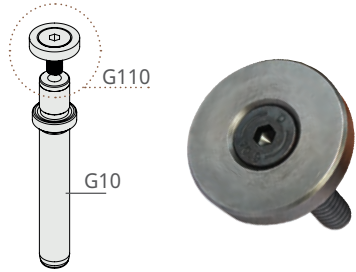
Flange Fitting: It is perpendicular angle locating on to die plate surface.

Operating Components: G100
G33- G34 - G35 - G36 - G40 - G42

Order: G15. d1 x L1

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Advantageous Product:
It is used as guiding component at dies and blow moulds.
Pillar mounting diameter and bush mounting diameter have the same tolerance value. In machining of guide pillar / bush holes on dies with the advantage of using same cutting tool (drill / reamer) for both guide pillar & bushes hole.
Thus, the necessity of using different cutting tools will disappear.

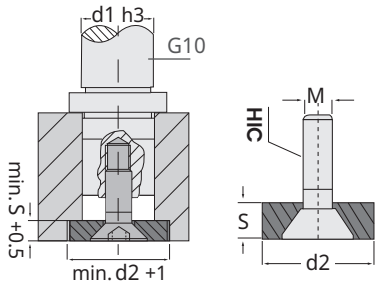


Mounting Flange

Code: G110

Guide pillar fixation & Support washer

- Reliable product used for guide pillar with centre collar mountings min. d3 + 1
- It can also be used to support for ejector plates in injection moulds / etc.



d2	d1 Pillar Diameter	S Flange	M Screw
22	Ø 15 / 16	5.5	M8 x 20
25	Ø 19 / 20		
32	Ø 24 / 25		
40	Ø 32 / 30	7.5	
50	Ø 38 / 40	9.5	
60	Ø 48 / 50		
73	Ø 60 / 63		
93	Ø 80	12	M12 x 20

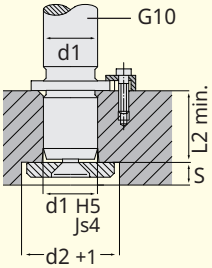
Order: G110. d2

Technical Info

Mounting Dimensions and Tolerances of Demountable Die Guide Pillars

Dimensions Given:
Recommended (suggestion) for G10 Guide pillar.

Mounting System:
Soft tight fitting system has been selected and determined for G10 Guide pillar.



d1	d1 H5	d2 +1	L2	S
15 / 16	+0.008	22	14.5	5.5
19 / 20	+0.009	25	17.5	5.5
24 / 25	+0.009	32	24.5	5.5
30 / 32	+0.011	40	29.5	7.5
38 / 40	+0.011	50	27.5	9.5
48 / 50	+0.013	60	37.5	9.5
60 / 63	+0.013	80	37.5	9.5
80	+0.013	95	48	12

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Guiding Components

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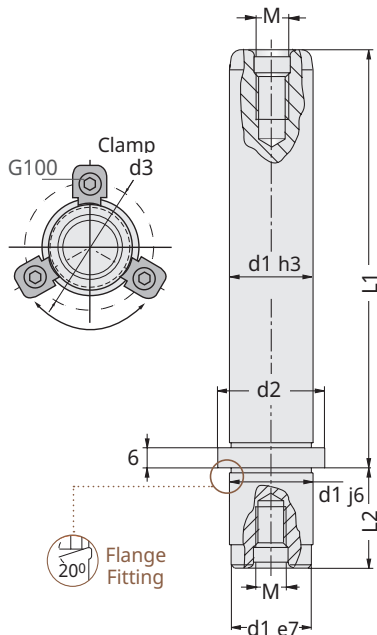
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Guiding Components

Code: **G10**

Guide Pillar with Centre Collar DIN 9825 / ISO 9182-5



Flange Fitting: It is perpendicular angle locating on to die plate surface.

Technical Info

It can be inserted directly to the holes which opened on the die plate. During mounting please avoid tight fitting. It can be positioned via soft-tight fitting to the holes and clamps or mounting flange with suitable tolerances or also with chemicals.

* Refer to the mounting measurement tolerance table on the **Page 63** for correct fitting of guide pillar to the die plate.

Please do not use two products without oil grooved together. One of the products should be with oil grooved.

Die Producers: Should pay attention to choose products suitable to the tolerances / standards in guide pillar & bush selection.

*To prevent incorrect closing of dies, three pieces main dimension and one piece auxiliary dimension can be used:
(15 - 19 - 24 - 32 - 38 - 48 - 60)*

For precision and safe mounting: G10 mounting flange.

d1	L1	L2	d2	d3 Clamp	M
15	100	20	22	33	M8
	112				
	125				
	140				
	160				
	180				
	200				
	220				
16	250	20	22	33	M8
	300				
	355				
	400				
	450				
	500				
	550				
	600				
19	100	23	25	36.4	M8
	112				
	125				
	140				
	160				
	180				
	200				
	220				
20	250	23	25	36.4	M8
	300				
	320				
	355				
	400				
	450				
	500				
	550				
24	100	30	32	44	M8
	112				
	125				
	140				
	160				
	180				
	200				
	224				
25	250	30	32	44	M8
	320				
	350				
	400				
	450				
	500				
	550				
	600				
32	112	37	40	52.4	M8
	125				
	140				
	160				
	180				
	200				
	224				
	250				
30	280	37	40	52.4	M8
	315				
	350				
	400				
	450				
	500				
	550				
	600				

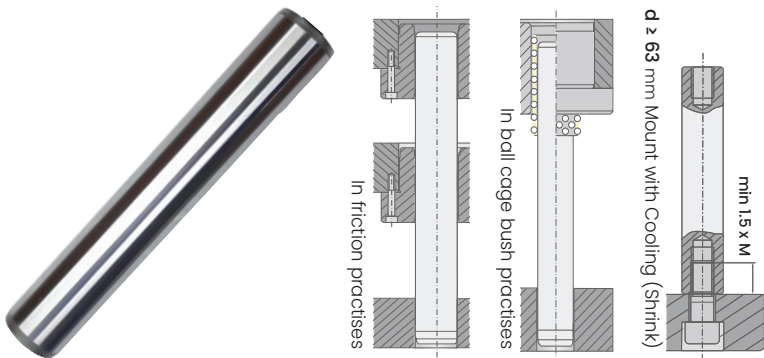
Operating Components:

All sliding systems also can be used with ball cage bush.

d1	L1	L2	d2	d3 Clamp	M
38	125	37	50	62.8	M8
	140				
	160				
	180				
	200				
	224				
	250				
	280				
40	315	37	50	62.8	M8
	355				
	400				
	450				
	500				
	550				
	600				
	650				
48	140	47	63	76	M8
	160				
	180				
	200				
	224				
	250				
	280				
	315				
50	355	47	63	76	M8
	400				
	450				
	500				
	550				
	600				
	650				
	700				
60	160	47	80	93.2	M8
	180				
	200				
	224				
	250				
	280				
	315				
	355				
63	400	47	80	93.2	M8
	450				
	500				
	550				
	600				
	650				
	700				
	750				
80	200	60	95	108.4	M 12
	224				
	250				
	280				
	315				
	355				
	400				
	450				

Order: **G10**. d1 x L1

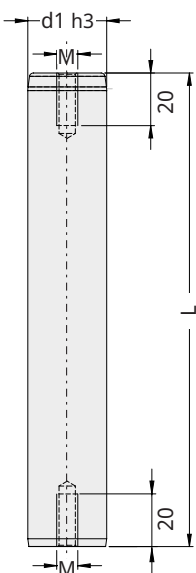
Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Guide Pillar, Plain & Threaded
DIN 9825 / ISO 9182

Code: **G24**

d1	L	M	d1	L	M
NEW	90	-		125	
10	100			140	
	112			160	
12	125			180	
	140			200	
15	90	M8	38	200	M8
	100			224	
	112		40	250	
	125			280	
	140			315	
16	160			355	
	180			400	
	200			140	M 12
19	100			160	
	112			180	
	125			200	
	140			224	
20	160		48	250	
	180			280	
	200		50	315	
	224			355	
	250			400	
24	100	M8		200	M 12
	112			224	
	125			250	
	140			280	
	160			315	
25	180		60	355	
	200			400	
	224		63	450	M 16
	250			500	
	280				
30	112				
	125				
	140				
	160				
32	180				
	200				
	224				
	250				
	280				
	315				



Technical Info
Section (d), two different products are to avoid incorrect closing of die during mounting, three pieces main dimension and one piece auxiliary should be used:
(d= 15 - 19 - 24 - 32 - 38 - 48 - 60 mm).
If extreme lateral forces are occurred at dies, in these cases self-lubricating wear plates should be used with pillars.

Order: G24. d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Operating Components:
All sliding systems also can be used with ball cage bush.

3D CAD FILES
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3D CAD - Formats: CATIA, STEP, IGES 3D, X_T
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Guide Pillar
DIN 9825 / ISO 9182

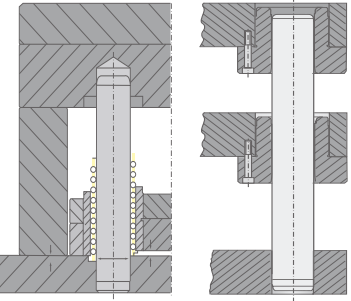
Code: **G19**

d1	L
12	100
	125
18	125
	160
30	160
	240

* In-die auxiliary guide pillar
* Guide pillars for ejector plate in injection moulds

Order: G19. d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Technical Info
It is used as auxiliary guiding component in dies or progressive die plates. It can be used with all sliding system or ball cage bush tools. When precision and iterative high speeds are required in injection mould ejector plates, it is suitable to use together with ball cage tools and also with sliding and self-lubricating guide bush components.

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Guiding Components

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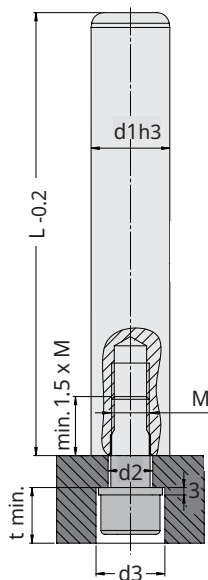
Guiding Components



Code: **G166**

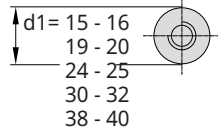
Guide Pillar - Bolted Type DIN 9825 / ISO 9182 - 2

G166 is supplied with a screw and washer.

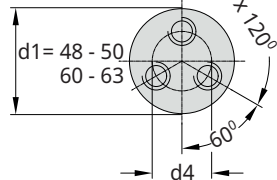


Hole pattern for pillar fastening:

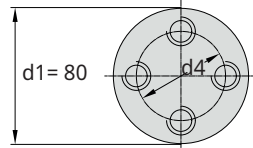
FORM - A



FORM - B



FORM - C



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3D CAD - Formats: CATIA, STEP, IGES 3D, X_T

d1	d2	d3	d4	t min.	M	L	Screw	Form
15 16	9	17	-	12	M8	90	M8 x 35	A
						100		
						112		
						125		
						140		
						160		
						180		
19 20	11	20	-	14	M10	100	M10 x 40	A
						112		
						125		
						140		
						160		
						180		
						200		
24 25	14	22	-	16	M12	100	M12 x 40	A
						112		
						125		
						140		
						160		
						180		
						200		
30 32	18	28	-	20.5	M16	224	M16 x 40	A
						250		
						280		
						315		
						355		
						315		
						355		
38 40	18	28	-	20.5	M16	160	M16 x 40	A
						180		
						200		
						224		
						250		
						280		
						315		
48 50	14	22	28	16	M12	180	M12 x 50	B
						200		
						224		
						250		
						280		
						315		
						355		
60 63	18	28	34	20.5	M16	250	M16 x 60	B
						280		
						315		
						355		
						400		
						450		
						500		
80	18	28	54	20.5	M16	280	M16 x 60	C
						315		
						355		
						400		
						450		
						500		
						550		

Technical Info

Section (d), two different products are to avoid incorrect closing of die during mounting, three pieces main dimension and one piece auxiliary should be used:
(d= 15 - 19 - 24 - 32 - 38 - 48 - 60 mm).

If extreme lateral forces are occurred at dies, in these cases self-lubricating wear plates should be used with pillars.



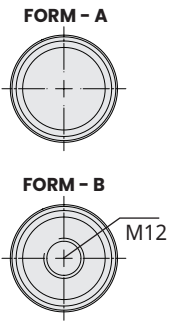
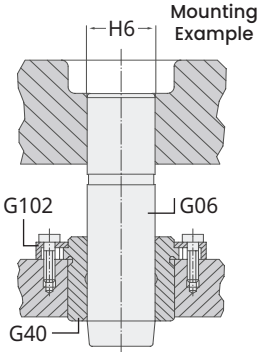
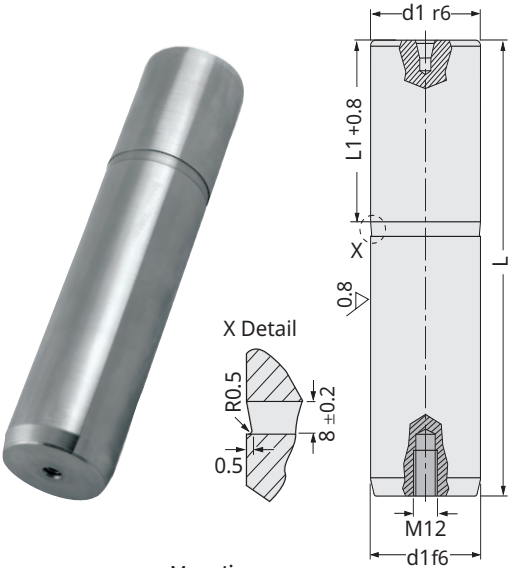
Order: **G166**. d1 x L



Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Operating Components:

All sliding systems also can be used with ball cage bush.



Guide Pillar / DIN 9833

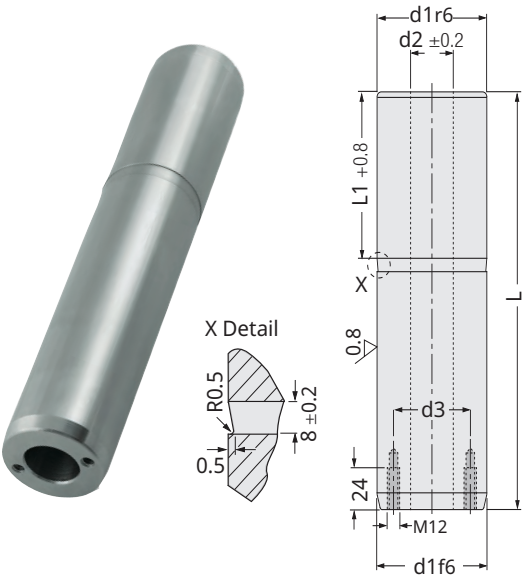
Code: **G06**

d1	L	L1	Form
32	160	45	A
	180		
	200		
40	200	56	A
	224		
	240		
50	224	70	A
	250		
	280		
63	250	80	A
	280		
	315		

d1	L	L1	Form
80	224	100	A
	250		
	280		
	315		
	355		
100	315	125	A
	355		
	400		
125	355	140	A
	400		
	450		
	500		
	550		

Order: **G06**. d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Guide Pillar / DIN 9833

Code: **G07**

d1	L	d2	d3	L1
80	250	40	58	100
	280			
	315			
	355			
100	315	50	72	125
	355			
	400			
125	355	65	90	140
	400			
	450			
	500			
	550			

Order: **G07**. d1 x L

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Customised Guide Units
Upon request, Güvenal manufactures highly complex and precise guide elements based on customer specifications and provides measurement protocols if needed. This includes both the reworking of catalogue products and the production of entirely new items.



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Guiding Components

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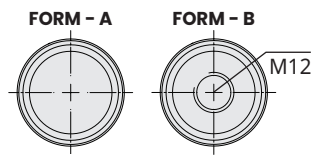
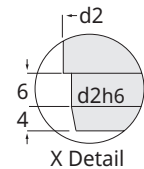
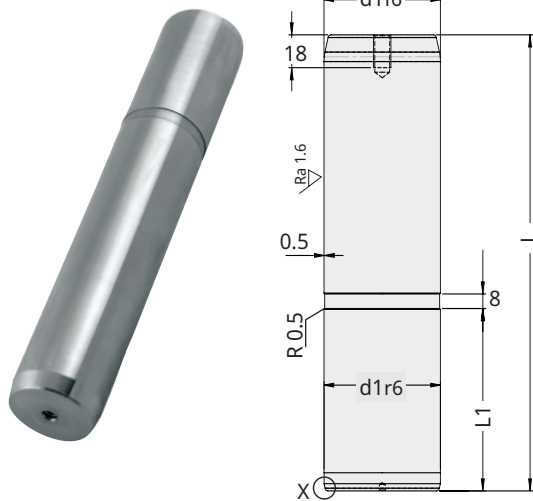
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Guiding Components

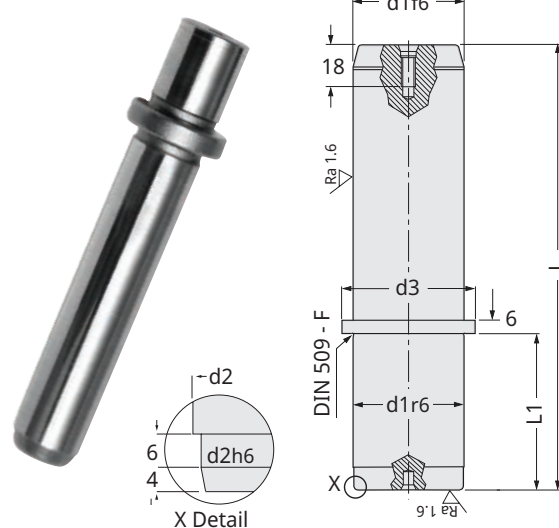


Guide Pillar Reference: WDX

Code: **G02**

d1	L	L1	Form
25	140	40	A
	160		
	180		
32	140	45	A
	160		
	180		
	200		
40	160	56	A
	180		
	200		
	224		
	250		
50	160	70	A
	180		
	200		
	224		
	250		
	280		
	315		

d1	L	L1	Form
25	220	40	A
	224		
	250		
	280		
32	280	80	A
	315		
	355		
	400		
40	250	100	B
	280		
	315		
	355		
	400		
50	315	125	B
	355		
	400		
	500		
	500		



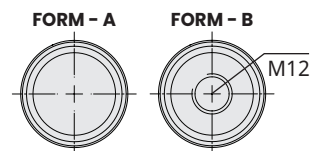
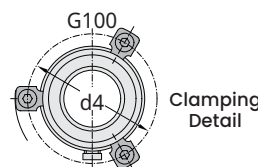
Guide Pillar with Centre Collar

Code: **G01**

Reference: WDX

d1	L	d3	L1	Form
25	140	33	40	A
	160			
	180			
32	140	40	45	A
	160			
	180			
	200			
40	160	50	56	A
	180			
	200			
	224			
	250			
50	160	60	70	A
	180			
	200			
	224			
	250			
	280			
	315			

d1	L	d3	L1	Form
25	220	33	40	A
	224			
	250			
	280			
32	280	40	45	A
	315			
	355			
	400			
40	250	50	56	A
	280			
	315			
	355			
	400			
50	250	60	70	A
	280			
	315			
	355			
	400			
63	280	80	80	A
	315			
	355			
	400			
80	250	90	100	B
	280			
	315			
	355			
	400			
100	315	110	125	B
	355			
	400			
	500			
	500			



d1	25	32	40	50	63	80	100
d4	45.2	52.4	62.8	73	93.2	103.4	123.4



Order:
G02. d1 x L



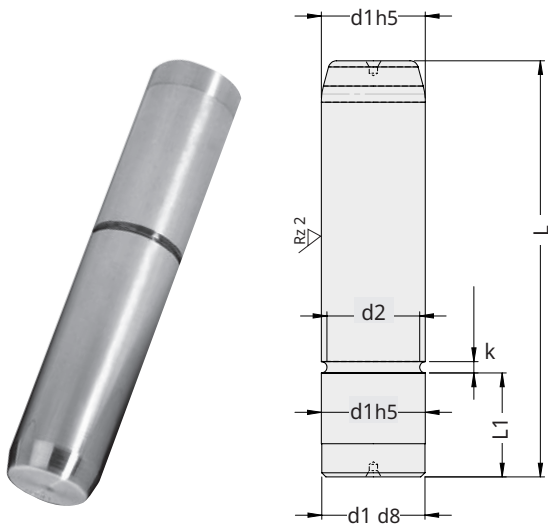
Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Order:
G01. d1 x L



Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Guide Pillar Reference: CNOMO Code: **G03**

Order Code	d1	L	d2	L1	k
G03 25100	25	100	22.3	25	2.7
G03 25125		125			
G03 25140		140			
G03 25160		160			
G03 25180		180			
G03 25200	32	200	27.8	32	4.2
G03 25224		224			
G03 32125		125			
G03 32140		140			
G03 32160		160			
G03 32180	40	180	35.8	63	4.2
G03 32200		200			
G03 32224		224			
G03 32250		250			
G03 40180		180			
G03 40200	40	200	35.8	63	4.2
G03 40224		224			
G03 40250		250			
G03 40280		280			
G03 40315		315			

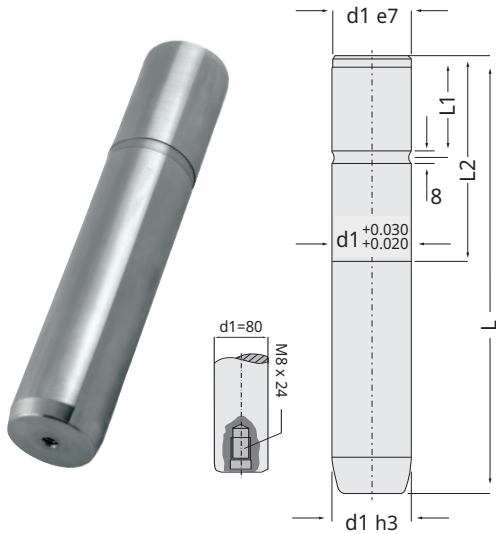


Material: 1.1213 (Cf53) / **Hardness:** 58 - 62 HRC

Order Code	d1	L	d2	L1	k
G03 50220	50	220	45.8	80	4.2
G03 50224		224			
G03 50250		250			
G03 50280		280			
G03 50315		315			
G03 50355	63	355	56.8	100	6.2
G03 50400		400			
G03 63250		250			
G03 63280		280			
G03 63315		315			
G03 63355	80	355	73.8	125	6.2
G03 63400		400			
G03 63450		450			
G03 63500	100	500	93.8	160	6.2
G03 80315		315			
G03 80355		355			
G03 80400	100	400	93.8	160	6.2
G03 80450		450			
G03 80500		500			
G03 100355	100	355	93.8	160	6.2
G03 100400		400			
G03 100450		450			
G03 100500		500			



For the compatibility chart of Guide Pillars and Bushes, please refer to the "Guiding Components – Technical Information" section at the beginning of Category 2.



Code: **G20**

Guide Pillar for Cast Block Bush

G20 Guide pillar type using with **G80** rectangular flange cast bush. In progressive form dies, it is reduced disassembly / assembly period of guide pillar & bush. No need to opening of guide pillar & bush holes in die is required. It reduces labor cost and also saves time by shortening production period of die. These are die components consisting of guide bush set inserted into lower and upper case.

d1	L	L1	L2	d1	L	L1	L2
40	180	37	77	63	240	60	120
	220				280		
	260				315		
	300				355		
	355				400		
	400				450		
50	200	47	95	80	240	60	120
	240				280		
	280				315		
	315				355		
	355				400		
	400				450		
	450				500		



Order: **G20**. d1 x L
Cast Bush: **G79**. d1

Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



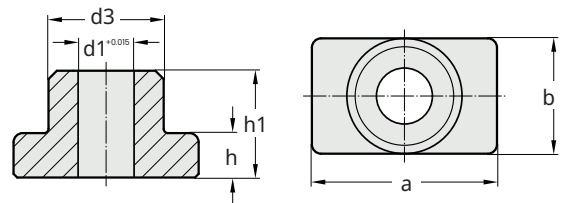
IMPORTANT:

Check perpendicularity of guide pillars after mounting.



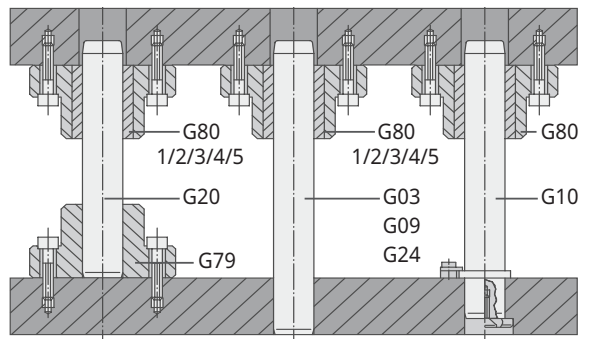
Code: **G79**

Flange Cast Bush - Rectangular



d1	d3	a	b	h	h1
40	80	130	80	30	77
50	96	160	96	35	95
63	110	180	110	35	120
80	130	215	130	40	120

Note: Other guide pillars conforming to rectangular flange cast bush connection. It can be used with **G03 - G09 - G24 - G10 - G20** in dies.



Flange Cast Bush Rectangular
Inner has been honed to $\varnothing d1 + 0.015$

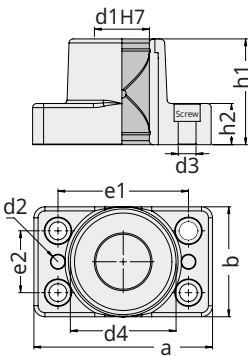
Dimension details of rectangular flange cast bush are same as other bushes (G80..)

Order: **G79**. d1

d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-1. d1**

Set Upon request
G20 + G79 + G80.1



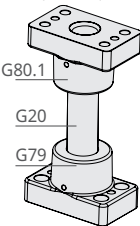
Flange Cast Bush Rectangular

Code: **G80 - 1**



Steel Type

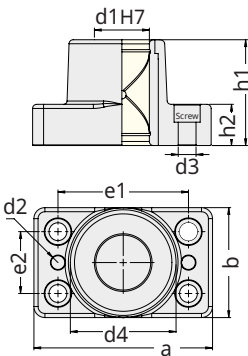
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-2. d1**

Set Upon request
G20 + G79 + G80.2



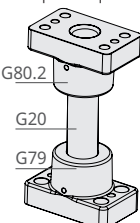
Flange Cast Bush Rectangular

Code: **G80 - 2**



Bronze Type

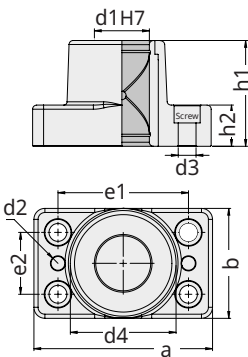
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-3. d1**

Set Upon request
G20 + G79 + G80.3



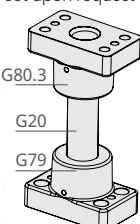
Flange Cast Bush Rectangular

Code: **G80 - 3**



Cast Type

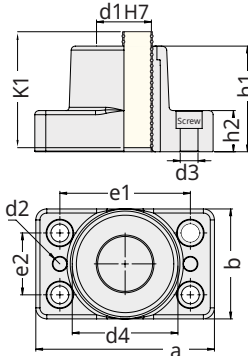
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	K1	Screw
40	130	80	77	30	10	13	70	95	45	80	12
50	160	96	95	35	10	15	83	118	55	105	14
63	180	110	120	35	12	17	94	132	62	140	16
80	215	130	120	40	14	21	112	160	75	140	20

Order: **G80-4. d1**

Set Upon request
G20 + G79 + G80.4 + G70



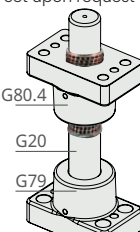
Flange Cast Bush Rectangular

Code: **G80 - 4**



Ball Cage Type

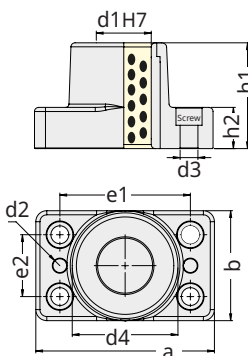
Set upon request



d1	a	b	h1	h2	d2	d3	d4	e1	e2	Screw
40	130	80	77	30	10	13	70	95	45	12
50	160	96	95	35	10	15	83	118	55	14
63	180	110	120	35	12	17	94	132	62	16
80	215	130	120	40	14	21	112	160	75	20

Order: **G80-5. d1**

Set Upon request
G20 + G79 + G80.5



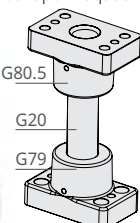
Flange Cast Bush Rectangular

Code: **G80 - 5**



Self-Lubricating Type

Set upon request



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Guiding Components

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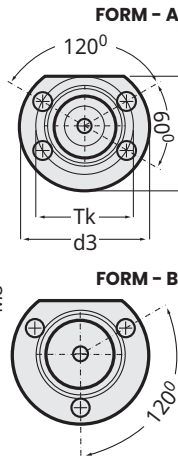
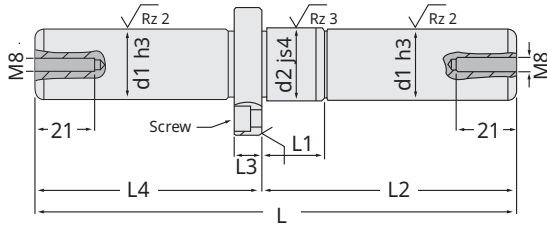
Guiding Components



Code: **G135**

Guide Pillar with Centre Flange

(Screw clamping guide pillar with centre collar)



d1	L	L1	L2	L4	L3	d2	d3	b	Tk	Screw	Form
20	150	22	80	70	10	22	44	39	33	M5 4 pcs.	A
	160		90	70							
	180		90	90							
	200		90	110							
25	170	22	90	80	10	26	50	45	38	M5 3 pcs.	B
	190		110	80							
	210		120	90							
	190		90	100							
	200		100	100							
	260		160	100							
30	220	27	100	120	10	32	58	53	46	M5 3 pcs.	B
	190		100	90							
	200		110	90							
	260		160	100							
	220		110	110							
	230		120	110							
	250		120	130							
32	260	27	130	130	10	34	60	55	48	M5 3 pcs.	B
	190		100	90							
	200		110	90							
	260		160	100							
	220		110	110							
	230		120	110							
	250		120	130							
40	260	27	130	130	12	42	70	65	56	M6 3 pcs.	B
	210		110	100							
	220		120	100							
	240		120	120							
	250		130	120							
50	270	36	130	140	15	52	80	75	66	M6 3 pcs.	B
	250		120	130							
	260		110	150							
	270		120	150							



Order: **G135**. d1 x L2 x L



Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Guide Pillar with Centre Flange

It can be used in standard pillar systems with many bearing bush connections. No clamping is necessary for die mounting position. It is fixed with hexagon socket head cap screws. The feature of centre flange: The protruding parts near the plate is prevented and a partial empty area is obtained in die mounting. It can be directly mounted in slots drilled on die plate. Shrink fit punchdown should not be done during mounting. It can be inserted with compatible tolerances with medium / shrink fit (bushes) and positioned with screws.

Compatible Bushes



G70



G151



G38



G37



G74



G75



G34



G33



G36



G35



G40

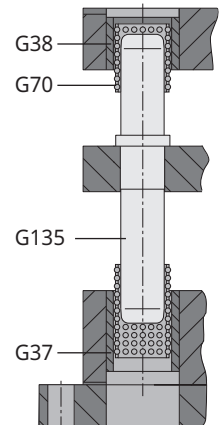
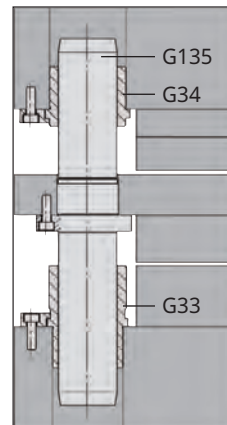


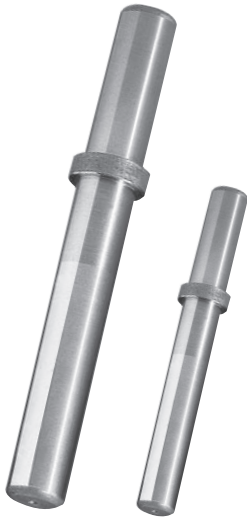
G41



Guide Pillar with Centre Flange (Ball cage bearing unit)

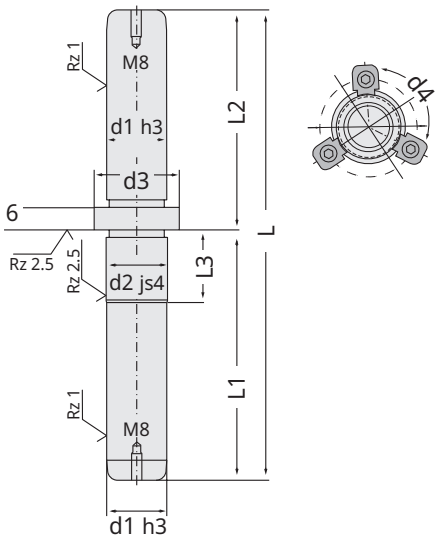
Dies which are designed with this ball cage bush have high speed and rigid connection. Hole tolerance for shrink fit is N5. Fixing place of die guide pillar affects securing lateral load resistance of die bushings. In dies with guide plates or having guide pillar mounted to die from bottom or top, if the distance (L) applying power is equal, bending and rotary values of guide pillars are equal. By securing die guide pillar to the guide plate there will be important improvements at bending values of pillars. Until the distance (L / 2) between application point of power and fastening surface reduces to half, load lifting capacity is increased 8 times.





Code: **G08**

Guide Pillar with Centre Fixing



Compatible Bushes



d1	L	L1	L2	L3	d2	d3	d4
20	160	90	70	19	21	26	37.6
	170	100					
	180		110				
	190	90					
	200						
	210						
25	180	100	80	22	26	32	44
	190	110					
	200		120				
	210	100					
	220						
	230						
30	180	100	80	25	31	38	50.4
	190	110					
	200		120				
	210	100					
	220						
	230	130					
	240		140				
	250						
40	200	110	90	27	41	50	62.8
	210	120					
	220		130				
	230	110					
	240						
	250						

 Order: **G08**. d1 x L

 **Material:** 1.1213 (Cf53)
Hardness: 58 – 62 HRC



Hole tolerance for shrink fit is N5. Fixing place of die guide pillar affects securing lateral load resistance of die bushings. In dies with guide plates or having guide pillar mounted to die from bottom or top, if the distance (L) applying power is equal, bending and rotary values of guide pillars are equal. By securing die guide pillar to the guide plate, there will be important improvements at bending values of pillars. Until the distance ($L / 2$) between application point of power and fastening surface reduces to half load lifting capacity is increased 8 times.

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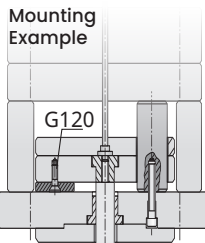
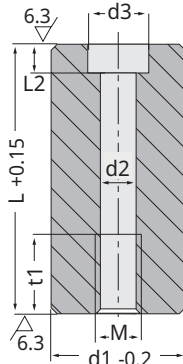
Guiding Components



Support Pillar for Injection Mould (Ejector plate / thrust wedge)

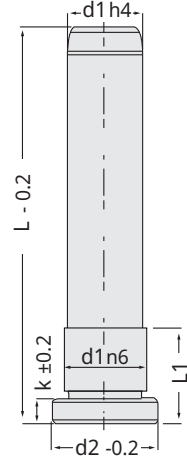
Code: **G21**

d1	L	d2	d3	L2	t1	M
32	47	6.5	11	7.5	15	M8
	57					
	67					
	77					
	87					
40	97					
	47	8.5	15	10	20	M10
	57					
	67					
	77					
50	87					
	97					
	117					
	47	8.5	15	10	20	M10
	57					
63	67					
	77					
	87					
	97					
	117					
80	137	10.5	18	12	25	M12
	157					
	57					
	67					
	77					
	87					
	97					
	117					
	137					
	157					



In injection moulds;
The thrust wedge that can be used in order to avoid dent load between support plate (H4) and bottom joint plate (H5A) also can be provided working of ejector plates more rigidly and sensitively. In order to avoid clicking (gap) **G110 mounting flange** can be used.

Order:
G21. d1 x L



Guide Pillar for Ball Cage

Code: **G18**

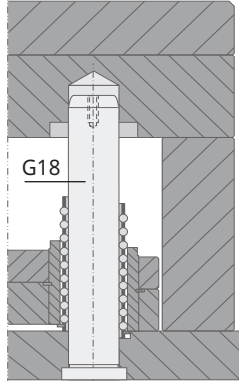
It is used in ejector plates of injection moulds and ejector plates with ball cage bush working precision and serially. In addition, it is compatible to work with intermediate plate of progressive dies / dies as auxiliary centering component.

d1	L	L1	d2	k
10	80	12	13	4
	100			
	120			
12	80	17	16	4
	100			
	120			
18	120	22	22	6
	140			
	160			
20	120	22	24	6
	140			
	160			
25	200	27	28	6
	240			
	280			
30	160	36	36	6
	200			
	240			
40	180	45	48	10
	200			
	240			

Compatible Bushes



Mounting Example



Order:
G18. d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

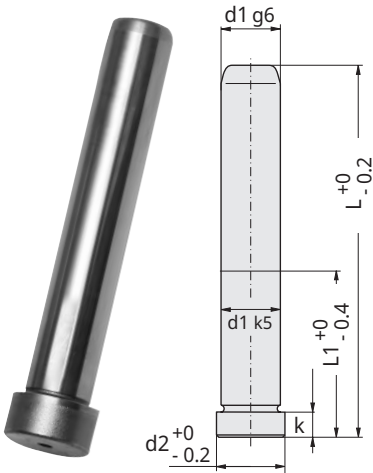
G18 - Areas of Usage:

- * In ejector plates, ball cage bush precision ejector systems.
- * It is used in dies as intermediate centering components.



Download
www.guvenal.net

3D CAD - Formats:
CATIA, STEP, IGES 3D, X_T



Guide Pillar
(without oil groove)

Code: **G17**

d1	L	L1	d2	k
10	40	13	12	3
	60			
	80			
	100	15		
	120			
12	40	18	16	6
	60			
	80			
	100			
	120	22		
	140			
	160			
14	60	22	18	8
	80			
	100			
	120			
	140	26		
	160			
	180			

d1	L	L1	d2	k
16	40	16	20	8
	60	24		
	80			
	100			
	120			
	140	26		
	160			
	180			
	200			
18	60	26	22	8
	80			
	100			
	120			
	140	28		
	160			
	180			
	200			
	240			
20	60	28	24	8
	80			
	100			
	120			
	140	32		
	160			
	180			
	200			
	240			
22	80	37	26	15
	100			
	120			
	140	43		
	160			
	180			
	200			
	220			
	240			
	300			

d1	L	L1	d2	k
24	80	39	28	15
	100			
	120			
	140	45		
	160			
	180			
	200			
	220			
	240			
	300			
25	80	35	28	15
	100			
	120			
	140			
	160	44		
	180			
	200			
	220			
	240			
	300			
30	100	45	36	15
	120			
	140			
	160			
	180	53		
	200			
	220			
	240			
	300			
	40			
180				
200				
220		63		
240				
280				
300				
320				
360				
50	160	65	58	15
	200			
	240			
	300			
	360			

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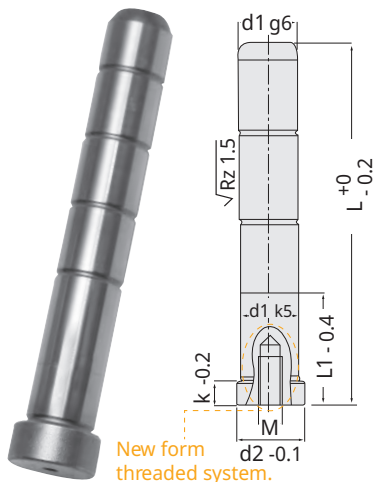
Guiding Components

Order: **G17**. d1 x L

Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

Operating Components: with all lubricated type bushes in injection moulds /dies.





Guide Pillar (with oil groove)

Code: **G11**

It can be fixed by thread in cap direction. By disassembling mould / die without dismantling, it provides process iteration advantage. The corners at oil grooves are chamfered (it reduces friction).

d1	L	L1	d2	k	M
10	40	13	12	3	-
	60				
	80				
	100	15			
	120				
12	40	18	16	6	-
	60				
	80				
	100				
	120	22			
	140				
	160				
14	60	22	18	8	-
	80				
	100				
	120				
	140	26			
	160				
	180				

d1	L	L1	d2	k	M
16	40	16	20	8	6
	60	24			
	80				
	100				
	120				
	140	26			
	160				
	180				
	200				
	220				
	250				
	300				
18	60	26	22	8	8
	80				
	100				
	120				
	140	28			
	160				
	180				
	200				
	240				
20	60	28	24	8	8
	80				
	100				
	120				
	140				
	160	32			
	180				
	200				
	220				
	240				
	250				
	300				
320					
22	80	37	26	15	8
	100				
	120				
	140	43			
	160				
	180				
	200				
	220				
	240				
	300				

d1	L	L1	d2	k	M
24	80	39	28	15	8
	100				
	120				
	140	45			
	160				
	180				
	200				
	220				
	240				
	300				
25	80	35	29	10	8
	100				
	120				
	140				
	160	44			
	180				
	200				
	220				
	250				
	300				
	320				
	350				
30	100	45	36	15	8
	120				
	140				
	160				
	180	53			
	200				
	220				
	240				
	300				
	360				
400					
40	160	55	48	15	8
	180				
	200				
	220	63			
	240				
	280				
	300				
	320				
	360				
50	160	65	58	15	8
	200				
	240				
	300	73			
	360				

IMPORTANT!

Injection Mould: Our guide pillars & bushes cover standard mould systems. But, guide pillar & bushes with different tolerances should be selected for aluminum injection moulds (high temperature / pressure). It should be requested from our company (production with special materials).

ATTENTION !

Please do not use two products together that have no oil grooves. One of these products should be oil grooved.



Order: **G11**. d1 x L



Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

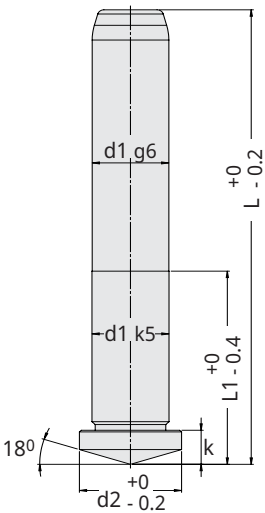
Operating Components:

With all unlubricated steel bushes in injection moulds or dies.

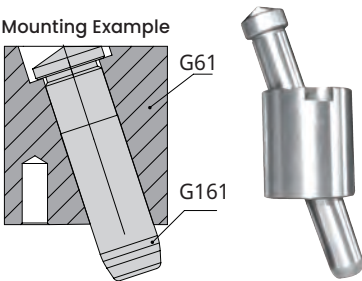


Angled Guide Pillar **Code: G161**

- Compatible Guide Bush: "G61"
- No reworking necessary before mounting.



Mounting Example



Order: **G161**. d1 x L

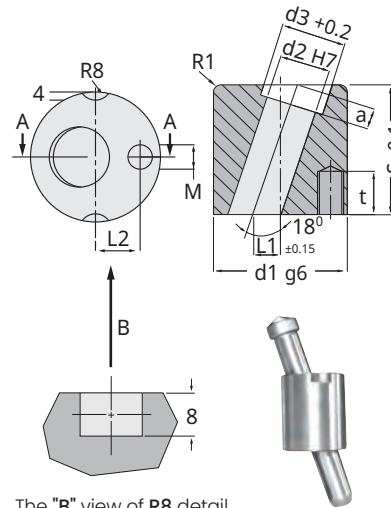


Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

d1	L	L1	d2	k
12	60	34	16	6
	80			
	100			
	120			
	140			
	160			
14	60	34	18	8
	80			
	100			
	120			
	140			
	160			
16	60	43	20	8
	80			
	100			
	120			
	140			
	160			
20	60	43	24	8
	80			
	100			
	120			
	140			
	160			
25	60	52	28	15
	80			
	100			
	120			
	140			
	160			
30	60	60	36	15
	80			
	100			
	120			
	140			
	160			



Angled Guide Bush **Code: G61**
(Core pillar holder)



The "B" view of R8 detail.

d2	s	d1	a	d3	L1	L2	M	t
12	36	32	6	17	7	10	M6	11
14	36	36	8	19	7	12	M8	16
16	46	40	8	21	8.5	13.5	M8	16
20	46	45	8	25	8.5	15	M8	16
25	56	50	15	29	10	16	M8	16
30	66	63	15	37	12	21	M10	20



Order: **G61**. d2



Material: 1.7131 (16MnCr5)
Hardness: 58 - 62 HRC



This unit eliminates angular perforation requirement. To insert this unit, only a hole is perforated at the front of the mould.
Angled guide bush can be used with **G161, G14 or G17** guide pillar.

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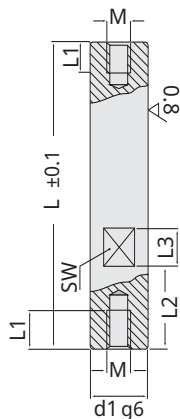
15

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Guiding Components



Ejector Rod

Code: **G16**

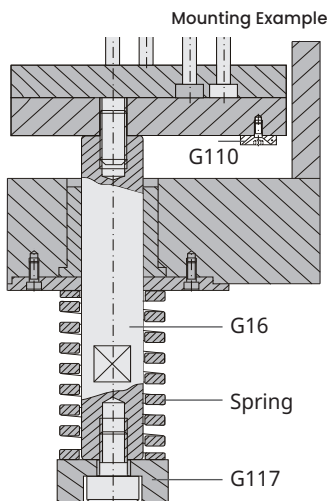
d1	L	M	L1	L2	L3	SW
10	60	M6	10	21	10	8
	70					
	80					
	100					
	140					
14	60	M8	13	21	10	10
	80					
	100					
	120					
	140					
16	60	M10	16	21	10	12
	70					
	80					
	100					
	120					
18	100	M10	20	30	10	14
	120					
	140					
	160					
	180					



Order: **G16**. d1 x L

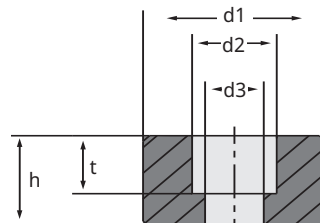
Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 – 62 HRC

d1	L	M	L1	L2	L3	SW
20	120	M12	24	40	16	14
	140					
	160					
	180					
	200					
24	120	M12	24	40	16	17
	140					
	160					
	180					
	200					
34	160	M16	34	50	16	24
	200					
	240					
	260					
	300					



Code: **G117**

Mounting Washer for Ejector Rod



Hex-Socket Head Cap Screw

d1	h	t	d2	d3
15	10	8.4	11	6.6
22	14	11	14.5	9
36	20	13.5	17.5	11
46	25	15.5	20	14
56	32	21	26	18



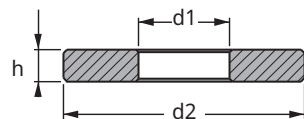
Order: **G117**. d1 x h



Code: **G114**

Thick Mounting Washer

Mould inner mounting washer / thick



Form	d1	d2	h	weight
M6	6.4	17	3	6 gr.
M8	8.4	23	4	10 gr.
M10	10.5	28	4	16 gr.
M12	13	35	5	32 gr.
M14	15	40	5	40 gr.
M16	17	45	6	60 gr.
M18	19	45	6	60 gr.
M20	21	50	6	75 gr.



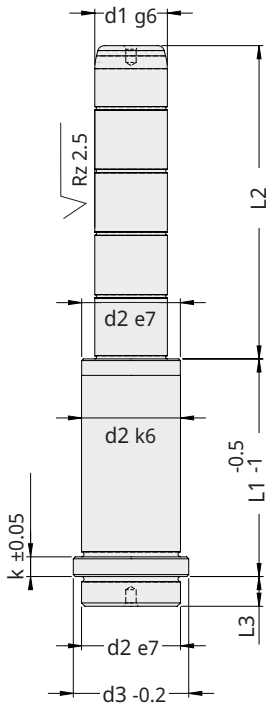
Order: **G114**. Form



Code: **G13D**
(without oil groove)
Upon request

Code: **G13**

Guide Pillar with Centring Collar
(with oil groove & without oil groove)



Code: **G13** (with oil groove)
Guide Pillar with Centring Collar



Code: **G13D** (without oil groove)
Guide Pillar with Centring Collar

d1	L1	L2	L3	d2	d3	k
15	17	35	9	20	25	6
14		55				
		75				
		95				
15	22	30	9	20	25	6
14		50				
		70				
		90				
		110				
		125				
	27	150				
15		30	9	20	25	6
14		45				
		65				
		85				
		105				
	36	125				
15		145				
14		165				
		35	9	20	25	6
		55				
		75				
		95				
		115				
		125				
	46	145				
15		155				
14		35	9	20	25	6
		45				
		65				
		85				
		105				
	56	125				
15		145				
14		165				
		185				

d1	L1	L2	L3	d2	d3	k
15	66	55	9	20	25	6
14		65				
		85				
		95				
	76	105	9	20	25	6
15		125				
14		35				
		55				
		75				
		95				
	86	125	9	20	25	6
15		55				
14		75				
		95				
	96	55	9	20	25	6
15		75				
14		95				
	116	55	9	20	25	6
15		75				
14		95				
	17	35	9	26	31	6
18		55				
20		75				
		120				
	22	35	9	26	31	6
18		45				
20		65				
		85				
	27	95	9	26	31	6
18		115				
20		35				
		45				
		65				
		85				
	56	105				
15		125				
14		145				
		165				

Compatible Bushes



Note:

To avoid wrong closing of mould holder plates, 2 different products are used.

Main dimensions: 14 - 20 - 24 - 30 - 40
other auxiliary dimensions:
15 - 18 - 22 - 32 - 42.

To avoid wrong closing of mould holder plates, can be used at 3 main dimensions guide pillar and 1 auxiliary dimension guide pillars. Our mould pillars were polished with surface polishing machine at final stage of production. The products with oil groove are available in our stocks.



Order: **G13**. d1 x L1 x L2
(with oil groove)



Order: **G13D**. d1 x L1 x L2
(without oil groove)



Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Continued on the
next page →

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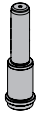
17

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Guiding Components



Code: G13 (with oil groove)
Guide Pillar with Centring Collar



Code: G13D (without oil groove)
Guide Pillar with Centring Collar

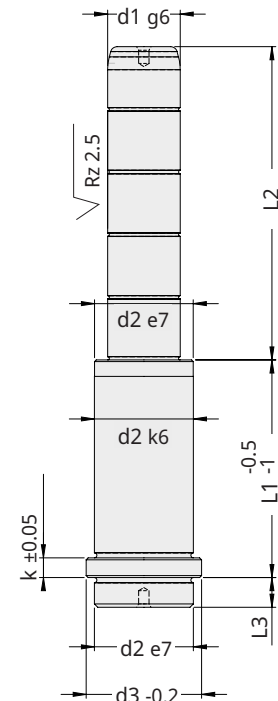
d1	L1	L2	L3	d2	d3	k
32 30	56	45	9	42	47	6
		55				
		75				
		95				
		115				
		135				
		155				
		175				
		245				
32 30	66	45	9	42	47	6
		55				
		75				
		95				
		115				
		135				
		155				
		175				
		245				
32 30	76	55	9	42	47	6
		75				
		95				
		115				
		135				
		155				
		225				
		245				
		295				
32 30	86	55	9	42	47	6
		75				
		95				
		115				
		135				
		155				
		225				
		245				
		295				
32 30	96	55	9	42	47	6
		75				
		95				
		115				
		135				
		155				
		205				
		245				
		295				
32 30	106	75	9	42	47	6
		95				
		115				
		125				
		145				
		165				
		245				
		295				
		345				

d1	L1	L2	L3	d2	d3	k
32 30	116	75	9	42	47	6
		95				
		115				
		135				
		155				
32 30	136	95	9	42	47	6
		115				
		135				
		155				
		175				
32 30	156	115	9	42	47	6
		155				
		175				
32 30	196	155	9	42	47	6
		195				
		215				
42 40	46	95	12	54	60	10
		165				
42 40	56	75	12	54	60	10
		115				
		155				
		195				
42 40	66	75	12	54	60	10
		135				
42 40	76	75	12	54	60	10
		115				
		175				
42 40	86	75	12	54	60	10
		135				
42 40	96	75	12	54	60	10
		115				
		155				
42 40	116	95	12	54	60	10
		135				
		195				
42 40	136	95	12	54	60	10
		135				
		215				
42 40	156	115	12	54	60	10
		155				
		215				
42 40	196	155	12	54	60	10
		195				
		235				
42 40	246	165	12	54	60	10
		245				



Code: G13D
(without oil groove)
Upon request

Code: G13
Guide Pillar with Centring Collar
(with oil groove & without oil groove)



Order: **G13**. d1 x L1 x L2
(with oil groove)



Order: **G13D**. d1 x L1 x L2
(without oil groove)



Material: 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Note: To avoid wrong closing of mould holder plates, 2 different products are used.

Main dimensions: 14 - 20 - 24 - 30 - 40 other **auxiliary dimensions:** 15 - 18 - 22 - 32 - 42. To avoid wrong closing of mould holder plates, can be used at 3 main dimensions guide pillar and 1 auxiliary dimension guide pillars. Our mould pillars were polished with surface polishing machine at final stage of production. The products with oil groove are available in our stocks.



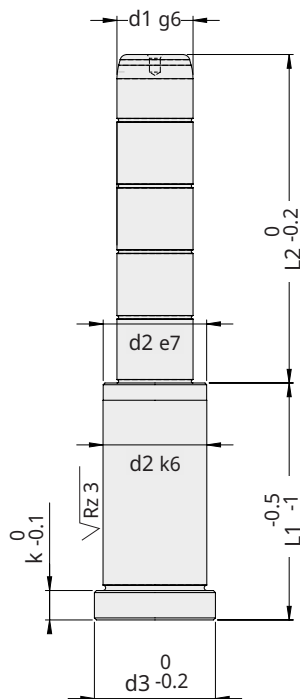
for Turkish Mould
Standard



Code: **G12D**
(without oil groove)
Upon request

Code: **G12**

Guide Pillar – Shouldered
(with oil groove & without oil groove)



Order: **G12**. d1 x L1 x L2
(with oil groove)



Order: **G12D**. d1 x L1 x L2
(without oil groove)



Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 – 62 HRC



Continued on the
next page →

d1	L1	L2	d2	d3	k
12	26	26	16	19	4
		36			
		46			
		56			
12	36	26	16	19	4
		36			
		46			
		56			
12	46	26	16	19	4
		36			
		46			
		56			
14	26	26	20	25	6
		36			
		46			
		56			
14	36	26	20	25	6
		36			
		46			
		56			
14	46	26	20	25	6
		36			
		46			
		56			
14	56	26	20	25	6
		36			
		46			
		56			
16	26	26	24	28	6
		36			
		46			
		56			
16	36	26	24	28	6
		36			
		46			
		56			
16	46	26	24	28	6
		36			
		46			
		56			
16	56	26	24	28	6
		36			
		46			
		56			
18	26	26	26	31	6
		36			
		46			
		56			
18	36	26	26	31	6
		36			
		46			
		56			
18	46	26	26	31	6
		36			
		46			
		56			
18	56	26	26	31	6
		36			
		46			
		56			
20	26	26	28	32	6
		36			
		46			
		56			
20	36	26	28	32	6
		36			
		46			
		56			
20	46	26	28	32	6
		36			
		46			
		56			
20	56	26	28	32	6
		36			
		46			
		56			

d1	L1	L2	d2	d3	k
16	56	26	24	28	6
		36			
		46			
		56			
16	66	26	24	28	6
		36			
		46			
		56			
16	76	26	24	28	6
		36			
		46			
		56			
16	86	26	24	28	6
		36			
		46			
		56			
18	26	26	26	31	6
		36			
		46			
		56			
18	36	26	26	31	6
		36			
		46			
		56			
18	46	26	26	31	6
		36			
		46			
		56			
18	56	26	26	31	6
		36			
		46			
		56			
20	26	26	28	32	6
		36			
		46			
		56			
20	36	26	28	32	6
		36			
		46			
		56			
20	46	26	28	32	6
		36			
		46			
		56			
20	56	26	28	32	6
		36			
		46			
		56			

d1	L1	L2	d2	d3	k
18	56	26	26	31	6
		36			
		46			
		56			
18	66	26	26	31	6
		36			
		46			
		56			
18	76	26	26	31	6
		36			
		46			
		56			
18	86	26	26	31	6
		36			
		46			
		56			
20	26	26	28	32	6
		36			
		46			
		56			
20	36	26	28	32	6
		36			
		46			
		56			
20	46	26	28	32	6
		36			
		46			
		56			
20	56	26	28	32	6
		36			
		46			
		56			

Compatible Bushes



IMPORTANT !

Injection Mould: Our guide pillars & bushes cover standard mould systems. But, guide pillar & bushes with different tolerances should be selected for aluminum injection moulds (high temperature / pressure). It should be requested from our company (production with special materials).



Code: **G12** (with oil groove)
Guide Pillar – Shouldered



Code: **G12D** (without oil groove)
Guide Pillar – Shouldered



for Turkish Mould
Standard

d1	L1	L2	d2	d3	k
20	66	26	28	32	6
		36			
		46			
		56			
		66			
		76			
20	76	96	28	32	6
		116			
		26			
		36			
		46			
		56			
20	86	66	28	32	6
		76			
		86			
		106			
		126			
		26			
20	96	36	28	32	6
		46			
		56			
		66			
		76			
		86			
20	96	106	28	32	6
		126			
		26			
		36			
		46			
		56			
25	26	66	34	38	8
		76			
		96			
		26			
		36			
		46			
25	36	56	34	38	8
		66			
		76			
		86			
		106			
		26			
25	46	36	34	38	8
		46			
		56			
		66			
		76			
		96			
25	56	116	34	38	8
		26			
		36			
		46			
		56			
		66			
25	56	76	34	38	8
		96			
		116			
		26			
		36			
		46			

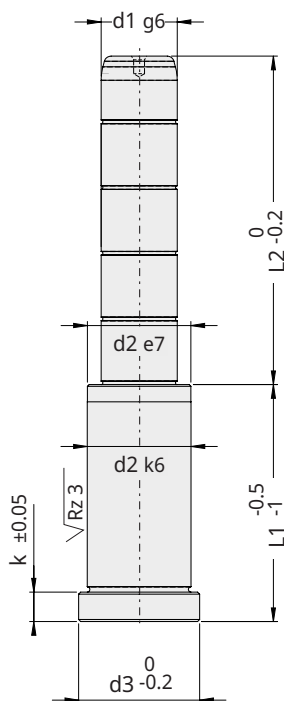
The tolerances and working space of our products are determined according to **DIN 7161**.



Code: **G50D**
(without oil groove)
Upon request

Code: **G50**

Guide Pillar – Shouldered
(with oil groove & without oil groove)



Order: **G50**. d1 x L1 x L2
(with oil groove)



Order: **G50D**. d1 x L1 x L2
(without oil groove)



Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC



Continued on the
next page →

d1	L1	L2	d2	d3	k
14	17	35	20	25	6
		55			
		75			
		95			
14	22	20	20	25	6
		35			
		40			
		45			
		50			
		55			
		65			
		70			
		90			
14	27	110	20	25	6
		20			
		35			
		40			
		45			
		55			
		65			
		85			
		105			
14	36	20	20	25	6
		35			
		40			
		45			
		55			
		65			
		75			
		95			
		105			
14	46	20	20	25	6
		35			
		45			
		65			
		85			
		105			
		20			
		35			
		55			
14	56	75	20	25	6
		95			
		55			
		65			
		95			
		20			
		35			
		55			
		95			
14	66	55	20	25	6
		65			
		95			
		20			
14	76	35	20	25	6
		45			
		95			
		20			
14	86	55	20	25	6
		65			
		95			
		20			
14	96	55	20	25	6
		65			
		95			
		20			
14	116	75	20	25	6
		95			
		20			
		35			

d1	L1	L2	d2	d3	k
15	17	35	20	25	6
		55			
		75			
		95			
15	22	20	20	25	6
		35			
		40			
		45			
		50			
		55			
		65			
		70			
		90			
15	27	110	20	25	6
		20			
		35			
		40			
		45			
		55			
		65			
		85			
		105			
15	36	20	20	25	6
		35			
		40			
		45			
		55			
		65			
		75			
		95			
		105			
15	46	20	20	25	6
		35			
		45			
		65			
		85			
		105			
		20			
		35			
		55			
15	56	75	20	25	6
		95			
		55			
		65			
		95			
		20			
		35			
		55			
		95			
15	66	55	20	25	6
		65			
		95			
		20			
15	76	35	20	25	6
		45			
		95			
		20			
15	86	55	20	25	6
		65			
		95			
		20			
15	96	55	20	25	6
		65			
		95			
		20			
15	116	75	20	25	6
		95			
		20			
		35			

d1	L1	L2	d2	d3	k
18	17	35	26	31	6
		55			
		75			
		95			
18	22	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
18	27	80	26	31	6
		85			
		115			
		20			
		35			
		40			
		45			
		50			
		55			
18	36	60	26	31	6
		65			
		70			
		80			
		85			
		105			
		125			
		20			
		35			
18	46	40	26	31	6
		45			
		50			
		55			
		60			
		65			
		70			
		75			
		80			
18	56	95	26	31	6
		115			
		135			
		20			
		45			
		65			
		85			
		105			
		135			
18	66	55	26	31	6
		65			
		95			
		20			
18	76	35	26	31	6
		45			
		95			
		20			
18	86	55	26	31	



Code: **G50** (with oil groove)
Guide Pillar – Shouldered



Code: **G50D** (without oil groove)
Guide Pillar – Shouldered



Continued on the next page →

d1	L1	L2	d2	d3	k
18	56	20	26	31	6
		35			
		55			
		75			
		95			
18	66	55	26	31	6
		75			
		95			
18	76	55	26	31	6
		75			
		95			
18	86	55	26	31	6
		75			
		95			
18	96	55	26	31	6
		75			
		95			
18	116	115	26	31	6
18	136	135	26	31	6
20	17	35	26	31	6
		55			
		75			
		95			
20	22	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		80			
		85			
		115			
20	27	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		80			
		85			
		105			
20	36	20	26	31	6
		35			
		40			
		45			
		50			
		55			
		60			
		65			
		70			
		75			
		80			
		95			
20	36	115	26	31	6
		135			
		135			
		135			
		135			
		135			
		135			
		135			
		135			
		135			
		135			
		135			

d1	L1	L2	d2	d3	k
20	46	20	26	31	6
		45			
		65			
		85			
		105			
		135			
		165			
20	56	20	26	31	6
		35			
		55			
		75			
		95			
20	66	55	26	31	6
		75			
		95			
		95			
20	76	55	26	31	6
		75			
		95			
20	86	55	26	31	6
		75			
		95			
20	96	55	26	31	6
		75			
		95			
20	116	115	26	31	6
20	136	135	26	31	6
22	17	35	30	35	6
		55			
		75			
22	22	35	30	35	6
		55			
		75			
		105			
		130			
22	27	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			
		165			
22	36	25	30	35	6
		45			
		50			
		55			
		60			
		70			
		75			
		80			
		95			
		115			
		135			
		165			

d1	L1	L2	d2	d3	k
22	46	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			
		165			
22	56	25	30	35	6
		45			
		55			
		75			
		95			
22	66	55	30	35	6
		75			
		95			
		95			
22	76	25	30	35	6
		45			
		55			
		75			
		95			
22	86	55	30	35	6
		75			
		95			
22	96	55	30	35	6
		75			
		95			
22	116	115	30	35	6
		155			
		155			
22	136	135	30	35	6
22	156	155	30	35	6
24	17	35	30	35	6
		55			
		75			
24	22	35	30	35	6
		55			
		75			
		105			
		130			
24	27	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		75			
		80			
		85			
		105			
		125			
		165			

d1	L1	L2	d2	d3	k
24	36	25	30	35	6
		45			
		50			
		55			
		60			
		70			
		75			
		80			
		95			
		115			
		135			
165					
24	46	25	30	35	6
		45			
		50			
		60			
		65			
		70			
		80			
		85			
		105			
		125			
		165			
24	56	25	30	35	6
		45			
		55			
		75			
		95			
		115			
		165			
24	66	55	30	35	6
		75			
		95			
24	76	25	30	35	6
		45			
		55			
		75			
		95			
115					
24	86	55	30	35	6
		75			
		95			
24	96	55	30	35	6
		75			
		95			
24	116	75	30	35	6
		115			
		155			
24	136	135	30	35	6
24	156	155	30	35	6

The tolerances and working space of our products are determined according to DIN 7161.



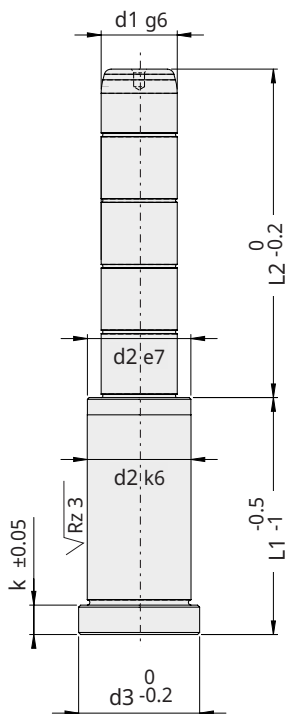
Code: **G50D**

(without oil groove)
Upon request

Code: **G50**

Guide Pillar – Shouldered

(with oil groove & without oil groove)



Order: **G50**. d1 x L1 x L2
(with oil groove)



Order: **G50D**. d1 x L1 x L2
(without oil groove)



Material: < Ø 20 = 1.7131
> Ø 20 = 1.1213 (Cf53)
Hardness: 58 - 62 HRC

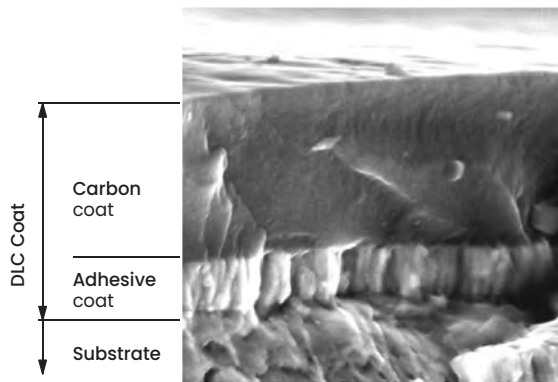
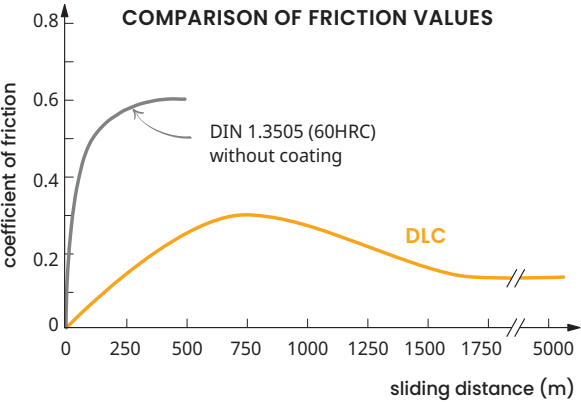


To be continued on the
previous page

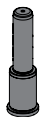
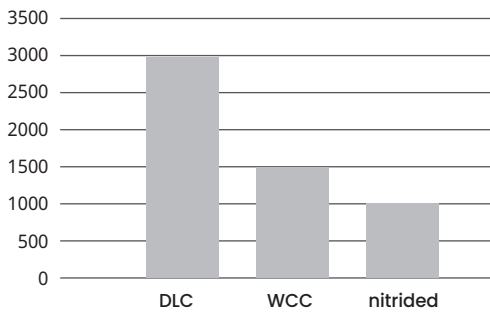
d1	L1	L2	d2	d3	k
30	22	35	42	47	6
		75			
		130			
30	27	45	42	47	6
		65			
		105			
		165			
30	36	55	42	47	6
		75			
		95			
		115			
		155			
30	46	45	42	47	6
		65			
		85			
		105			
		125			
30	56	165	42	47	6
		55			
		75			
		95			
		115			
30	66	135	42	47	6
		175			
		55			
		75			
		95			
30	76	115	42	47	6
		135			
		155			
		55			
		75			
30	86	95	42	47	6
		115			
		155			
		55			
		75			
30	96	95	42	47	6
		115			
		155			
		55			
		75			
30	116	95	42	47	6
		115			
		155			
		55			
		75			
30	136	115	42	47	6
		155			
		55			
		75			
		95			
30	156	115	42	47	6
		155			
		55			
		75			
		95			
30	176	135	42	47	6
		175			
		155			
		55			
		75			
30	196	155	42	47	6
		195			
		35			
		75			
		130			

d1	L1	L2	d2	d3	k
32	27	45	42	47	6
		65			
		105			
		165			
32	36	55	42	47	6
		75			
		95			
		115			
		155			
32	46	45	42	47	6
		65			
		85			
		105			
		125			
32	56	165	42	47	6
		55			
		75			
		95			
		115			
32	66	135	42	47	6
		175			
		55			
		75			
		95			
32	76	115	42	47	6
		135			
		155			
		55			
		75			
32	86	95	42	47	6
		115			
		155			
		55			
		75			
32	96	95	42	47	6
		115			
		155			
		55			
		75			
32	116	95	42	47	6
		115			
		155			
		55			
		75			
32	136	115	42	47	6
		155			
		55			
		75			
		95			
32	156	115	42	47	6
		155			
		55			
		75			
		95			
32	176	135	42	47	6
		175			
		155			
		55			
		75			
32	196	155	42	47	6
		195			
		35			
		75			
		135			

d1	L1	L2	d2	d3	k
40	76	75	54	60	10
		115			
		175			
40	86	75	54	60	10
		135			
40	96	75	54	60	10
		115			
		155			
40	116	95	54	60	10
		135			
		195			
40	136	95	54	60	10
		135			
		215			
40	156	115	54	60	10
		155			
		215			
40	176	135	54	60	10
		155			
		175			
40	196	155	54	60	10
		195			
		235			
42	36	75	54	60	10
		135			
42	46	95	54	60	10
		165			
42	56	75	54	60	10
		115			
		155			
42	66	195	54	60	10
		135			
42	76	75	54	60	10
		115			
		175			
42	86	75	54	60	10
		135			
42	96	75	54	60	10
		115			
		155			
42	116	95	54	60	10
		135			
		195			
42	136	95	54	60	10
		135			
		215			
42	156	115	54	60	10
		155			
		215			
42	176	135	54	60	10
		155			
		175			
42	196	155	54	60	10
		195			
		235			
50	96	115	66	72	10
	116	135			
	136	135			
	156	155			
	196	175			
60	96	115	80	86	10
	116	135			
	136	135			
	156	155			
	196	175			
	246	195			



Surface hardness HV



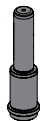
Code: **G50DLC** (DLC Coated)

Guide Pillar – Shouldered



Code: **G13DLC** (DLC Coated)

Guide Pillar with Centring Collar



Customised Guide Units

Upon request, Güvenal manufactures highly complex and precise guide elements based on customer specifications and provides measurement protocols if needed. This includes both the reworking of catalogue products and the production of entirely new items.

NEW

DLC Coated Guide Pillars



Code: **G13DLC**

Code: **G50DLC**

What is DLC ?

DLC means: Diamond like carbon.

The DLC coating is a sliding coating applied using the PACVD (Plasma-Enhanced Chemical Vapor Deposition) process. The coating is deposited through a physical and chemical reaction at approximately 180°C, with carbon-containing carrier gases introduced simultaneously. Due to the low coating temperature, DLC is also ideal for materials with a low tempering temperature. It combines excellent hardness with superior tribological properties in terms of friction, anti-adhesion, wear, and lubrication.

DLC coating is especially beneficial for applications involving motion or friction in mould making, such as ejection, guiding, and centering. With the DLC coating, it is possible to operate completely lubricant-free..

Advantages:

- Lubricant-free operation
- Extended maintenance intervals
- Very high abrasion resistance
- No seizing, even when running dry
- Good corrosion resistance

Technical Data:

Coating hardness: 2400 – 3000 HV

Coating thickness: ~ 1.5 – 2 µm

Coating temperature: 180°C

Temperature resistance: 350°C



DLC coated high-quality standard components to ensure a longer service life for your moulds.

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Guiding Components

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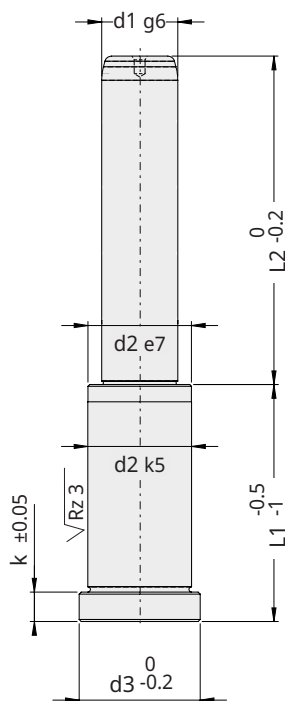
18

NEW



Guide Pillar - Shouldered

Code: **G50DLC** (DLC Coated)



Order:
G50DLC. d1 x L1 x L2



Material: 1.7131
Hardness: 60 HRC

Technical Data:

Coating hardness: 2400 - 3000 HV
Coating thickness: ~ 1.5 µm
Coating temperature: 180°C
Temperature resistance: 350°C

DLC coated high-quality standard components to ensure a longer service life for your moulds.

d1	L1	L2	d2	d3	k
14	22	35	20	25	6
		55			
		70			
		90			
14	27	35	20	25	6
		55			
		85			
		105			
14	36	35	20	25	6
		55			
		75			
		95			
14	46	35	20	25	6
		65			
		85			
		55			
14	56	55	20	25	6
		75			
14	66	55	20	25	6
		65			
		95			
18	27	35	26	31	6
		55			
		65			
		70			
		80			
		85			
		105			
		125			
18	36	35	26	31	6
		55			
		75			
		95			
		115			
		135			
		45			
18	46	65	26	31	6
		85			
		105			
		125			
18	56	55	26	31	6
		75			
		95			
		55			
18	66	75	26	31	6
		95			
		55			

d1	L1	L2	d2	d3	k
18	76	55	26	31	6
		75			
		95			
22	36	25	30	35	6
		45			
		55			
		75			
		95			
22	46	115	30	35	6
		135			
		165			
		25			
		45			
22	56	55	30	35	6
		75			
		95			
		115			
		165			
22	66	55	30	35	6
		75			
		95			
22	76	55	30	35	6
		75			
		95			
		115			
22	86	55	30	35	6
		75			
22	96	55	30	35	6
		75			
30	46	105	42	47	6
		125			
		165			
		25			
		45			

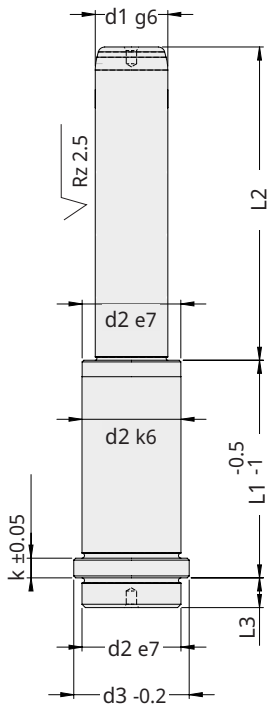
d1	L1	L2	d2	d3	k
30	56	55	42	47	6
		75			
		95			
		115			
30	66	135	42	47	6
		175			
		55			
30	76	75	42	47	6
		95			
		115			
		135			
30	86	175	42	47	6
		55			
		75			
		95			
30	96	115	42	47	6
		155			
		75			
		95			
30	116	115	42	47	6
		155			
30	136	95	42	47	6
		115			
30	156	115	42	47	6
		155			
40	96	75	54	60	10
		115			
40	116	135	54	60	10
		155			
40	136	95	54	60	10
		135			
40	156	215	54	60	10
		115			
40	196	155	54	60	10
		195			
		235			
		55			

NEW



Guide Pillar with Centring Collar

Code: **G13DLC** (DLC Coated)



Order:
G13DLC. d1 x L1 x L2

Material: 1.7131
Hardness: 60 HRC

Continued on the next page →

Technical Data:
Coating hardness: 2400 - 3000 HV
Coating thickness: ~ 1.5 µm
Coating temperature: 180°C
Temperature resistance: 350°C

d1	L1	L2	L3	d2	d3	k
14	22	30	9	20	25	6
		50				
		70				
		90				
14	27	30	9	20	25	6
		45				
		65				
		85				
14	36	105	9	20	25	6
		35				
		55				
		75				
14	46	95	9	20	25	6
		115				
		35				
		45				
14	56	65	9	20	25	6
		85				
		105				
		125				
14	66	35	9	20	25	6
		55				
		75				
		95				
14	76	125	9	20	25	6
		55				
		75				
		95				
20	22	35	9	26	31	6
		45				
		65				
		85				
20	27	35	9	26	31	6
		45				
		65				
		85				
20	36	105	9	26	31	6
		125				
		35				
		55				
20	46	35	9	26	31	6
		55				
		75				
		95				
20	56	115	9	26	31	6
		135				
		35				
		45				
20	66	35	9	26	31	6
		55				
		75				
		95				
20	76	105	9	26	31	6
		125				
		35				
		45				
20	86	35	9	26	31	6
		55				
		75				
		95				
20	96	125	9	26	31	6
		145				
		35				
		45				
24	27	65	9	30	35	6
		85				
		105				
		125				
24	36	165	9	30	35	6
		205				
		35				
		45				
24	46	65	9	30	35	6
		85				
		105				
		125				
24	56	165	9	30	35	6
		205				
		35				
		45				
24	66	65	9	30	35	6
		85				
		105				
		125				
24	76	135	9	30	35	6
		155				
		35				
		45				

d1	L1	L2	L3	d2	d3	k
20	56	35	9	26	31	6
		55				
		75				
		95				
20	66	115	9	26	31	6
		135				
		35				
		55				
20	76	65	9	26	31	6
		85				
		105				
		125				
20	86	145	9	26	31	6
		165				
		35				
		45				
20	96	65	9	26	31	6
		85				
		105				
		125				
24	27	165	9	30	35	6
		205				
		35				
		45				
24	36	65	9	30	35	6
		85				
		105				
		125				
24	46	165	9	30	35	6
		205				
		35				
		45				
24	56	65	9	30	35	6
		85				
		105				
		125				
24	66	135	9	30	35	6
		155				
		35				
		45				

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Guiding Components

Coating hardness: 2400 - 3000 HV
Coating thickness: ~ 1.5 µm

Coating temperature: 180°C
Temperature resistance: 350°C



Code: **G13DLC**

(DLC Coated)

Guide Pillar with Centring Collar

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Guiding Components

d1	L1	L2	L3	d2	d3	k
22	76	55	9	30	35	6
		75				
		95				
		115				
		135				
		155				
		55				
		75				
		95				
		115				
22	86	135	9	30	35	6
		155				
		195				
		55				
		75				
		95				
		115				
		135				
		155				
		195				
22	96	55	9	30	35	6
		75				
		95				
		115				
		135				
		155				
		195				
		55				
		75				
		95				
22	116	115	9	30	35	6
		135				
		155				
		195				
		55				
		75				
		95				
		115				
		135				
		155				
22	136	75	9	30	35	6
		95				
		115				
		55				
		75				
		95				
		115				
		135				
		155				
		195				
30	36	55	9	42	47	6
		75				
		95				
		115				
		135				
		155				
		195				
		55				
		75				
		95				
30	46	115	9	42	47	6
		135				
		155				
		195				
		235				
		55				
		75				
		95				
		115				
		135				
30	56	155	9	42	47	6
		195				
		235				
		55				
		75				

d1	L1	L2	L3	d2	d3	k
30	66	55	9	42	47	6
		75				
		95				
		115				
		135				
		155				
		195				
		235				
		55				
		75				
30	76	95	9	42	47	6
		115				
		135				
		155				
		195				
		235				
		55				
		75				
		95				
		115				
30	86	135	9	42	47	6
		155				
		195				
		235				
		55				
		75				
		95				
		115				
		135				
		155				
30	96	195	9	42	47	6
		235				
		55				
		75				
		95				
		115				
		135				
		155				
		195				
		235				
30	116	75	9	42	47	6
		95				
		115				
		135				
		155				
		195				
		235				
		55				
		75				
		95				
30	136	115	9	42	47	6
		135				
		155				
		195				
		235				
		55				
		75				
		95				
		115				
		135				
30	156	155	9	42	47	6
		195				
		235				
		55				
		75				
		95				
		115				
		135				
		155				
		195				
30	176	115	9	42	47	6
		135				
		155				
		195				
		235				
		55				
		75				
		95				
		115				
		135				

d1	L1	L2	L3	d2	d3	k
30	196	115	9	42	47	6
		155				
		195				
40	66	75	12	54	59	10
		95				
		115				
		155				
40	76	75	12	54	59	10
		95				
		115				
		155				
		195				
40	86	75	12	54	59	10
		95				
		115				
		155				
40	96	95	12	54	59	10
		115				
		155				
		195				
		235				
40	116	95	12	54	59	10
		115				
		155				
		195				
		235				
40	136	95	12	54	59	10
		115				
		155				
		195				
		235				
40	156	95	12	54	59	10
		115				
		155				
		195				
40	176	115	12	54	59	10
		155				
		195				
40	196	155	12	54	59	10
		195				
		235				

NEW



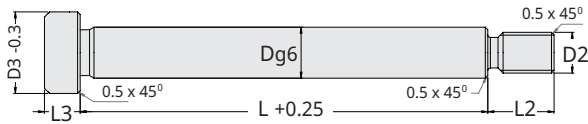
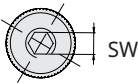
Tolerance
g6

Guiding Shoulder Screw

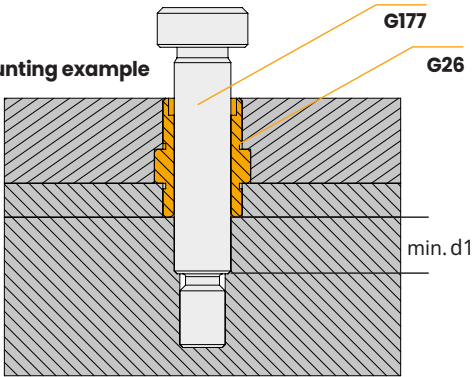
Code: **G177**

* Can be used as a guide bolt with stopper.

* **g6** tolerance



Mounting example



Order Code	D	L	D2	L2	D3	L3	SW
G177.10.25	10	25	M8	13	16	7	5
G177.10.30		30					
G177.10.40		40					
G177.10.50		50					
G177.10.60		60					
G177.10.70		70					
G177.10.80		80					
G177.10.90		90					
G177.10.100		100					
G177.12.25	12	25	M10	16	18	9	6
G177.12.30		30					
G177.12.40		40					
G177.12.50		50					
G177.12.60		60					
G177.12.70		70					
G177.12.80		80					
G177.12.90		90					
G177.12.100		100					
G177.12.120		120					
G177.14.30	14	30	M10	16	21	9	6
G177.14.40		40					
G177.14.50		50					
G177.14.60		60					
G177.14.70		70					
G177.14.80		80					
G177.14.90		90					
G177.14.100		100					
G177.14.120		120					
G177.14.140		140					
G177.18.40	18	40	M12	18	27	11	8
G177.18.50		50					
G177.18.60		60					
G177.18.70		70					
G177.18.80		80					
G177.18.90		90					
G177.18.100		100					
G177.18.120		120					
G177.18.140		140					

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Guiding Components